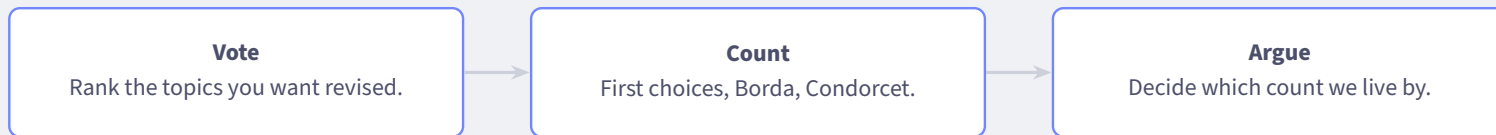


Social choice

One set of preferences, three winners



The prize is real: we revise the winner in class.

Three topics on the shortlist:

$A =$

$B =$

$C =$

How they ranked them	How many said it	First choice	Borda points

Borda points for three topics: 2 for a first place, 1 for a second, 0 for a third.

Most first choices

A

B

C

Winner

Borda score

A

B

C

Winner

Head to head

$A \vee B$

$A \vee C$

$B \vee C$

Condorcet winner

Did all three agree?

Counting first choices looks only at _____, and throws away _____.

Borda uses _____, so a topic that is everybody's second choice can beat one that half the room ranks last.

A Condorcet winner beats every other topic _____, and sometimes _____.

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So which method should we use for the class vote, and why?

We are revising:

What we just did in the room is **Question 1** on the Social Choice page, with nine students and three topics.

- (a) Rank by first choice votes and give the winner. [2]
- (b) The Borda scores and the Borda winner. [4]
- (c) Whether there is a Condorcet winner. [4]
- (d) Why the methods disagree, and which you would recommend. [5]

vknigh.org/gt/topics/social-choice.html